Emotion Unlocked: Multimodal Emotion Detection Through Audio-Video Fusion

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***Abstract*—Emotion recognition is an important area of human-computer interaction with a wide range of applications in healthcare, customer experience optimization, and sentiment analysis. This paper presents a multimodal deep learning architecture that combines both audio and video modalities to improve emotion classification accuracy. The audio model takes Mel-Frequency Cepstral Coefficients (MFCCs) as input features, fed into a deep learning pipeline consisting of Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Bidirectional Long Short-Term Memory (BiLSTM) layers to efficiently extract temporal dependencies from speech signals. In contrast, the video model utilizes a 3D Convolutional Neural Network (3D CNN) developed upon ResNet-18 to extract spatial-temporal features from video frames. The obtained audio and video feature representations are subsequently combined through a fully connected Multi-Layer Perceptron (MLP) that learns an optimal joint representation for final emotion classification. Through the exploitation of complementary information provided by each modality, the suggested multimodal strategy performs better than unimodal models, showing better emotion recognition accuracy and robustness on various cross-validation folds.**

***Index Terms*—Emotion Recognition, Multimodal Deep Learning, Audio-Visual Fusion, CNN, RNN, BiLSTM, 3D CNN, ResNet-18, MFCC, MLP, Human-Computer Interaction, Sentiment Analysis**

I. Introduction

Emotion recognition has emerged as an important research field in affective computing, artificial intelligence, and human-computer interaction [1]. The capability to accurately analyze and interpret human emotions from audio and video information has a wide range of applications in various fields. In monitoring mental health, emotion recognition can help diagnose diseases like depression and anxiety by analyzing speech and facial expressions of patients. Emotion recognition in human-robot interaction increases the empathy response of robots, which then makes them useful in caregiving, education, and customer support. Autonomous systems such as intelligent assistants and driver monitoring systems also gain from emotion recognition through increased user experience and safety. Security and surveillance see the detection of emotional states contribute to threat recognition and prevention of crime. Moreover, emotional recognition is also extensively used in entertainment, game development, and advertising to customize user experience and maximize content delivery [3].

Historical methods mainly used unimodal approaches that processed either voice or facial expressions separately [7]. However, human emotions are multimodal and involve a complex combination of vocal tone, facial actions, and other non-verbal signals. Deep learning has greatly enhanced emotion recognition with its capability for extracting subtle patterns from audiovisual data. Despite these advancements, challenges such as variations in emotional expression, data skewness, and real-time computation limitations still necessitate further exploration of more efficient fusion methods and robust learning algorithms [2] [6].

In this research, we aim to develop a multimodal emotion recognition system using deep learning methods. Specifically, we will use the RAVDESS dataset, which offers high-quality labeled audio-visual data, to train and test our models. Our approach will integrate feature extraction techniques for speech and facial expressions, utilizing deep learning architectures to improve recognition accuracy. By leveraging fusion strategies, we seek to combine the strengths of each modality in a way that enhances the overall robustness and adaptability of the system. This work contributes to the growing field of affective computing by developing more accurate emotion classification models and addressing existing challenges related to real-world application and computational efficiency.

Additionally, multimodal emotion recognition research is expanding beyond audiovisual analysis to incorporate physiological signals such as electroencephalography (EEG) and galvanic skin response (GSR) for a more comprehensive emotional understanding [15]. Advanced deep learning architectures like Adaptive Deep Belief Networks (DBNs) and multimodal data augmentation frameworks are being developed to improve accuracy and data efficiency [14]. These recent advancements provide the motivation for our research, where we focus on optimizing feature fusion techniques and leveraging deep learning for robust multimodal emotion detection.

II. Literature Survey

Emotion recognition has profoundly changed with the use of multimodal methods, using deep learning models and heterogeneous data for enhanced accuracy and resilience. In the early work, unimodal methods were mostly used, while contemporary studies focus on the fusion of audio and visual information for enhanced emotion classification [4].

Han et al. proposed EmoBed in 2019, a method that enhances monomodal emotion recognition by capitalizing on crossmodal emotion embeddings [11]. EmoBed makes use of joint multimodal training and crossmodal learning to capture complementary information in auxiliary modalities. The paper showed that even if only a single modality is present at inference time, EmoBed performs better than standard unimodal models by boosting feature representations learned at training.

In 2021, Wang et al. introduced an adversarial learning-based multimedia emotion tagging method [12]. Their framework is composed of an emotion classifier and a discriminator, where the classifier outputs emotion labels, and the discriminator provides statistical similarity between predicted and ground truth labels. The framework effectively captured emotion distributions and enhanced accuracy in multimedia emotion tagging tasks.

John and Kawanishi proposed a transformer-based multimodal emotion recognition model in 2022 [5]. Their model utilized self-attention and cross-attention mechanisms to improve feature extraction in both audio and video modalities. Tested on the RAVDESS, CREMA-D, and SAVEE datasets, their system performed better than traditional CNN-LSTM architectures. Their work emphasized the capability of transformer models to learn long-range dependencies in multimodal data.

In 2023, Deng and Ren created a Multi-label Emotion Detection Architecture (MEDA) [13]. MEDA uses a multi-channel emotion-specified feature extractor to separately process various emotions so that weaker emotions don't get masked by stronger ones. Additionally, the research proposed an emotion correlation learner in order to learn the dependencies between the emotions, which resulted in better accuracy while detecting co-occurring emotions.

Mocanu et al. (2023) introduced a cross-modal audio-video fusion technique that employed spatial, channel, and temporal attention mechanisms with deep metric learning [1]. Their model achieved better performance on the RAVDESS and CREMA-D datasets with accuracy rates of 89.25% and 84.57%, respectively. The authors highlighted the importance of attention-based feature extraction and cross-modal fusion in improving emotion classification. Their work also compared with state-of-the-art models and showed significant improvements.

Kumar and Martin (2023) contrasted several deep learning models for multimodal emotion recognition [8]. Their research compared several different fusion approaches, setting the supremacy of model-level fusion over decision- and feature-level fusion. Their work, tested on RAVDESS and IEMOCAP, showcased the prowess of deep learning models in achieving better multimodal emotion recognition accuracy. They also talked about challenges with real-time deployment and efficiency in computation and how to optimize deep learning models for large-scale deployment.

Kamada and Ichimura presented in 2024 a multimodal Adaptive Deep Belief Network (DBN) for emotional recognition with the RAVDESS dataset [14]. The model adaptively changes its network topology by creating or annihilating neurons to enhance learning efficiency. The paper emphasized the benefits of adaptive multimodal fusion by using different weight adaptations for audio and video streams, which reached 89.94% accuracy.

Li et al. (2025) addressed the problem of data shortages in multimodal emotion recognition by introducing a fusion-based data augmentation framework [15]. Their approach combines physiological signals with a ConvNeXt-Attention fusion model and uses a conditional self-attention GAN to synthesize more labeled data. The research showed that multimodal dataset augmentation can greatly enhance classification performance, with performance exceeding 95% on benchmark datasets.

Collectively, these studies verify that multimodal emotion recognition is superior to unimodal approaches. Through the use of attention mechanisms, transformer models, crossmodal fusion mechanisms, and data augmentation, researchers have been able to see significant improvements in accuracy. Yet, ongoing issues like domain adaptation, computational efficiency, dataset generalization, and deployment in the real world demonstrate the need for further progress towards pragmatic application [9] [10] .

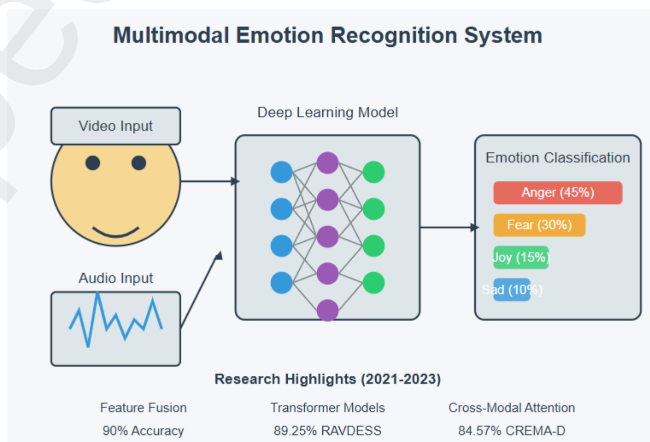

Fig. 1. Multimodal Emotion Recognition

III. Empirical Analysis

This section presents a detailed evaluation of different deep learning models employed for emotion recognition. The performance of each model is analyzed based on three approaches: audio-based models, video-based models, and fusion-based models. The results highlight the impact of different architectures and feature extraction techniques on classification accuracy.

*1) **Audio-Based Emotion Recognition**:* Emotion recognition from speech signals requires a model that can capture both spectral and temporal features. Various architectures were tested, combining convolutional and recurrent layers to improve feature extraction. The classification accuracies of different models for audio-based emotion recognition are given in Table I.

TABLE I
AUDIO-BASED MODEL ACCURACIES

Model	Accuracy (%)
CNN+LSTM+RNN	79
ResNet	75
CNN+LSTM	72
LSTM	68

The highest-performing model in this class is CNN+LSTM+RNN, with an accuracy of 79%. This combined model integrates convolutional layers (for spectral feature extraction) and LSTM/RNN layers (for temporal dependencies in speech signals). Adding RNN layers enhances sequential learning over CNN+LSTM alone.

The ResNet model, despite being optimized for image-based feature extraction, performed fairly well at 75%, implying that deep residual connections help learn high-level speech representations. The CNN+LSTM model fared slightly lower at 72%, showing that while CNN and LSTM combined capture spatial and temporal cues, the lack of additional recurrent layers limits performance. The LSTM-only model had the lowest accuracy (68%), proving that LSTMs alone struggle without CNN-based feature extraction.

*2) **Video-Based Emotion Recognition**:* Facial expressions play a crucial role in emotion recognition, making video-based models an effective method. To examine the influence of different video-processing architectures, we tested 3D CNNs and hybrid feature extraction approaches. The classification accuracies of video-based models are listed in Table II.

TABLE II
VIDEO-BASED MODEL ACCURACIES

Model	Accuracy (%)
Feature Extraction Using ResNet 3D + 3D CNN	92
3D CNN	72

The ResNet 3D + 3D CNN approach significantly outperformed the standalone 3D CNN, achieving 92% accuracy. This highlights the importance of pretrained feature extraction using deep residual networks before classification. ResNet 3D allows hierarchical feature extraction, capturing both spatial and temporal dependencies in facial expressions.

Conversely, the baseline 3D CNN model attained only 72% accuracy, showing that while 3D CNNs can process spatial and temporal features together, they may not generalize well without additional feature refinement. This suggests that larger datasets and optimization techniques are needed for training a 3D CNN from scratch.

*3) **Fusion-Based Emotion Recognition**:* Although unimodal methods (audio-only or video-only) are informative, combining both modalities is anticipated to enhance the accuracy and robustness of emotion recognition. Various fusion strategies were investigated, and their classification performances are listed below.

The best accuracy of 96% was obtained by the MLP-based fusion model, showing that fully connected deep learning models can successfully fuse audio and video feature representations. This shows that if both modalities are fused at the feature level and then fed into an MLP classifier, the model can learn intricate interdependencies, which results in better classification performance.

The Autoencoder + MLP model was close second at 91% accuracy. Autoencoders help in feature compression and dimensionality reduction, which might discard redundant information prior to classification. Yet the performance dip from the baseline MLP indicates that compression by autoencoders can occasionally eliminate the vital features used for proper emotion identification.

TABLE III
FUSION-BASED MODEL ACCURACIES

Model	Accuracy (%)
MLP	96
Autoencoder + MLP	91
Averaging Probabilities	85

The Averaging Probabilities method, which calculated the average of the predicted probabilities of the separate audio and video models, gave the lowest accuracy (85%). This shows that statistical fusion methods, which do not learn the interactions between the modalities, are inferior to deep learning-based fusion methods. The findings point towards learning-based fusion methods performing better than mere probability-based fusion methods by being able to model the intricate interactions between speech and facial expressions more effectively.

IV. PROPOSED METHODOLOGY

A. Data Preprocessing

*1) **Audio Preprocessing**:* Raw audio signals are then processed to yield useful features for emotion classification. Raw waveforms have redundant and irrelevant information, so Mel-Frequency Cepstral Coefficients (MFCCs) are calculated since they are capable of representing human auditory perception efficiently. To preserve the temporal pattern in speech, the signal is divided into overlapping short frames. A pre-emphasis filter is used to boost high-frequency components, and this increases the distinctiveness of speech features. The filtered signal is next converted to the frequency domain by means of the Short-Time Fourier Transform (STFT), represented by:

$$X(m,k) = \sum_{n=0}^{N-1} x(n)w(n)e^{-j2\pi kn/N} \tag{1}$$

where $w(n)$ represents the Hamming window function, which smooths the signal to reduce spectral leakage. The

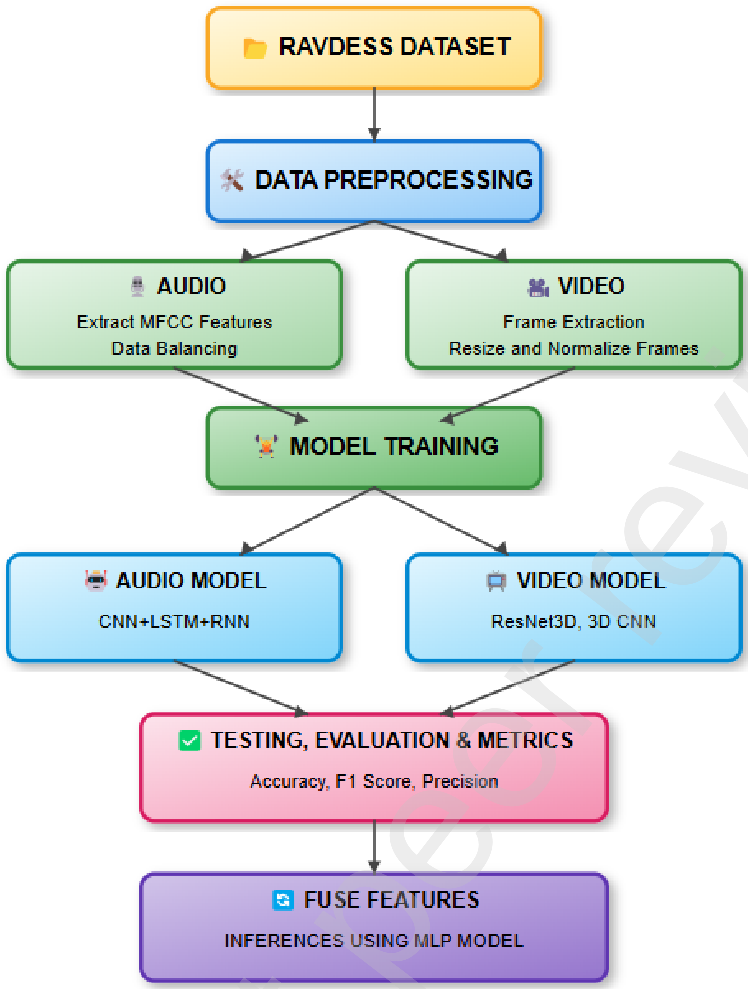

Fig. 2. Proposed Methodolgy

output is passed through a Mel-filter bank, which applies a triangular filter to simulate human auditory perception:

$$S(m) = \sum_{k=1}^{K} |X(m,k)|^2 H_k \quad (2)$$

where H_k is the triangular filter function. Finally, Dimensionality Reduction is performed using the Discrete Cosine Transform (DCT) to obtain a compact 40-dimensional MFCC feature vector, which serves as input to the deep learning model:

$$c_n = \sum_{m=1}^{M} \log(S(m)) \cos\left[n(m-0.5)\frac{\pi}{M}\right] \quad (3)$$

2) ***Video Preprocessing***: In order to make sure that facial expressions are captured well, the video clips are preprocessed before they are input into the deep learning model. The videos are sampled to 16 frames per sequence to ensure temporal consistency between samples. As irrelevant background features can add noise, facial detection and cropping are done using OpenCV to make sure that only expression-based information is preserved. The cropped frames are resized to 112×112 pixels and normalized with:

$$I' = \frac{I - \mu}{\sigma} \quad (4)$$

where μ and σ represent the mean and standard deviation of pixel intensities across the dataset. Finally, the processed frames are converted into tensors to enable efficient processing by the deep learning model.

B. Modality-Specific Models

1) ***Audio Model (CNN-BiLSTM-RNN)***: The extracted audio features are passed through a convolutional neural network (CNN) to learn spatial representations. The convolution operation is given by:

$$h_{i,j} = \sum_{p=1}^{P} w_p \cdot x_{i+p,j} + b \quad (5)$$

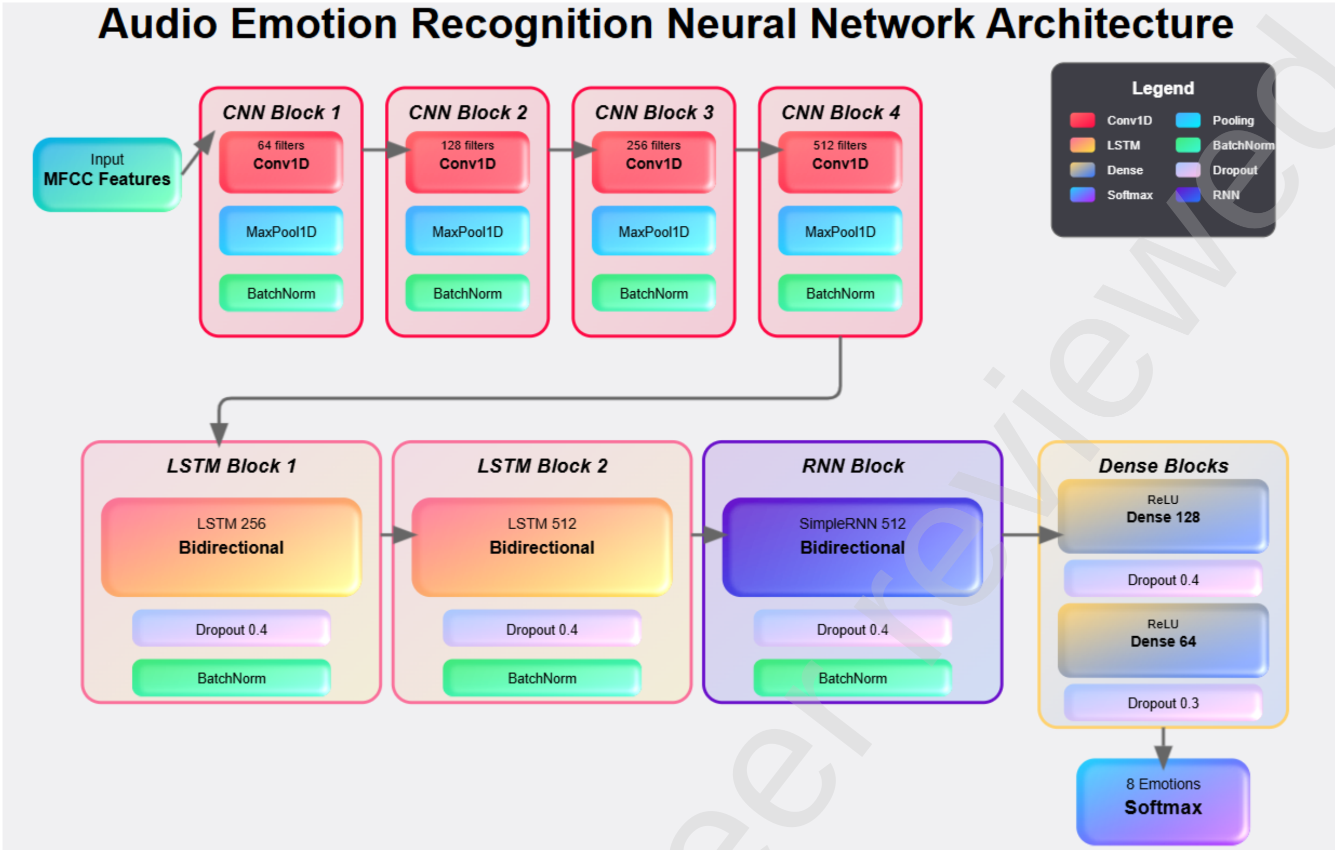

Fig. 3. Architecture of Audio Model

where w_p represents the convolution kernel weights, and $x_{i,j}$ is the input feature map. These extracted features are then fed into Bidirectional Long Short-Term Memory (BiLSTM) layers, which capture dependencies across time:

$$h_t = \sigma\left(W_h h_{t-1} + W_x x_t + b_h\right) \tag{6}$$

where h_t represents the hidden state at time step t, and W_h, W_x, and b_h are the trainable weight matrices and bias term. The final representation is obtained through fully connected layers, which map the features to a 128-dimensional vector, forming the final audio feature representation.

*2) **Video Model (3D CNN - ResNet-18)**:* The preprocessed video frames are input to a 3D Convolutional Neural Network (3D CNN) built on ResNet-18, which learns spatial and temporal patterns from facial expressions. Unlike 2D CNNs, which extract spatial features only, 3D CNNs perform convolutions in both spatial and temporal dimensions, making them more effective for video-based tasks. The 3D convolution operation is represented as

$$h_{i,j,k} = \sum_{p=1}^{P} \sum_{q=1}^{Q} \sum_{r=1}^{R} w_{p,q,r} \cdot x_{i+p,j+q,k+r} + b \tag{7}$$

where (P, Q, R) are the 3D kernel size, and $w_{p,q,r}$ are the learnable weights. The feature maps are then processed using Global Average Pooling (GAP) to perform dimensionality reduction while preserving vital information. The final 512-dimensional feature vector is extracted and sent into fully connected layers for further processing.

C. Feature Fusion

As emotions are expressed both in speech and facial expressions, the fusion of these modalities provides a more comprehensive and accurate emotion recognition model.

*1) **Fusion Mechanism**:* The fusion mechanism combines both feature-level information as well as modality-specific predictions of the audio and video models.

The end fused feature representation includes the following:

- 40-dimensional MFCC feature vector (audio) from speech.
- 8-dimensional probability vector (audio model output), the model's confidence for each of the 8 emotion classes.
- 512-dimensional feature vector (video) from facial expressions.
- 8-dimensional probability vector (video model output), the model's confidence in each of the 8 emotion classes.

*2) **Prediction-Based Fusion**:* Aside from feature-level fusion, the model also fuses predictions at the emotion probability distribution level using the output of the audio and

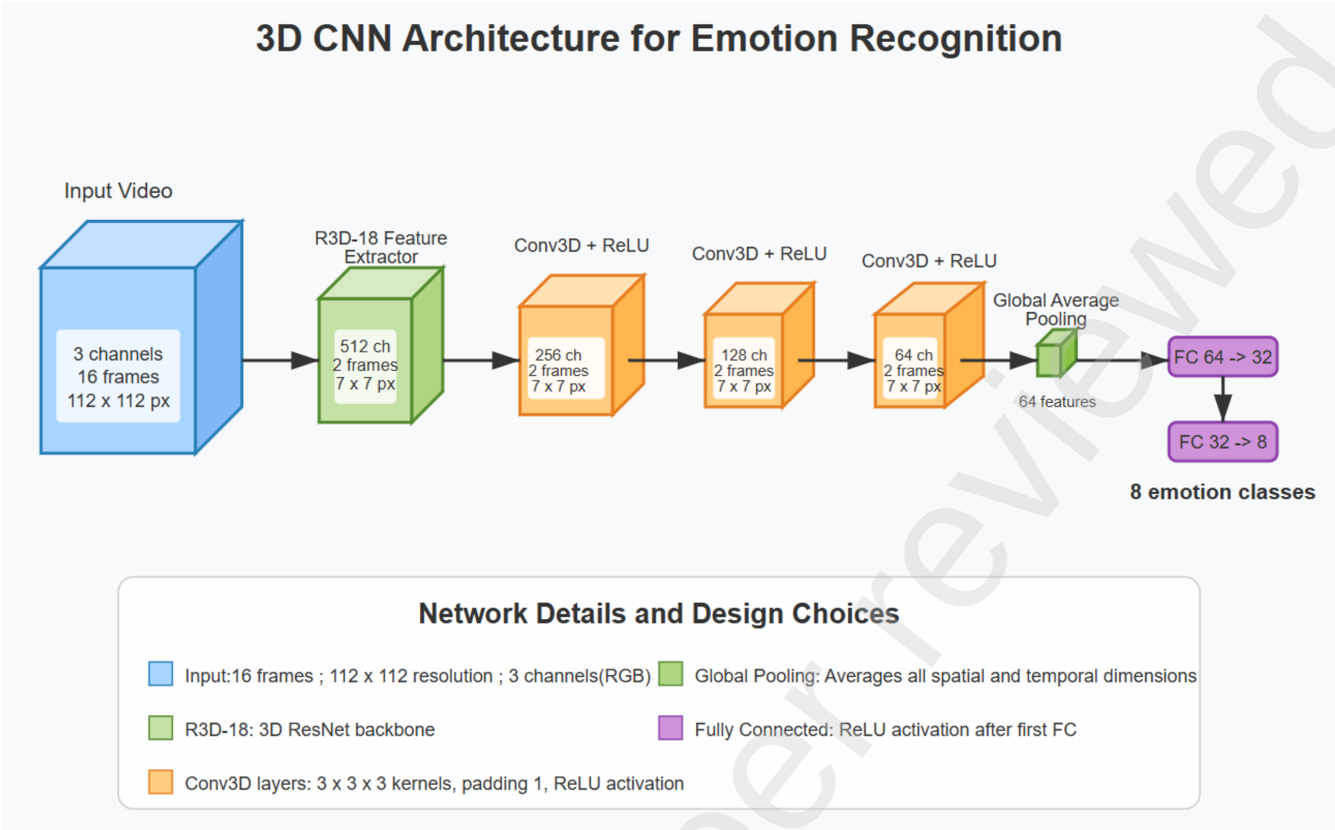

Fig. 4. Architecture of Video Model

video models. Each model outputs Softmax probabilities that correspond to its confidence in each of the 8 emotion classes.

3) ***Final Classification Using MLP:*** A Multi-Layer Perceptron (MLP) classifier processes the 568-dimensional fused representation, applying nonlinear transformations to extract deeper representations of emotion.

D. Model Training

1) ***Loss Function:*** The model is trained with the Categorical Cross-Entropy (CCE) loss function:

$$L = -\sum_{i=1}^{N} y_i \log(\hat{y}_i) \tag{8}$$

where:

- y_i represents the true class label (one-hot encoded).
- $\hat{y}_i$ is the forecasted probability for class i.
- N is the total number of classes (i.e., 8 for our model).

To address possible class imbalances in the dataset, the model uses a weighted CCE loss function. As some emotions might be more frequent than others, greater weightage is given to minority classes so that the model won't be biased against the majority classes. The weighted loss is calculated as:

$$L = -\sum_{i=1}^{N} w_i y_i \log(\hat{y}_i) \tag{9}$$

where w_i represents the class weight assigned to each emotion category.

This class-weighting strategy ensures that the model learns equally from all emotions, even if some are less frequent in the dataset.

2) ***Optimization Algorithm:*** The model is trained with the Adam optimizer, a gradient-based adaptive learning technique that adjusts the learning rate dynamically from past gradients and momentum. As opposed to fixed learning rate standard Stochastic Gradient Descent (SGD), Adam updates weights to achieve faster convergence and stability by making use of:

$$\theta_t = \theta_{t-1} - \frac{\alpha}{\sqrt{v_t} + \epsilon} m_t \tag{10}$$

where:

- θ_t is the model parameters at time step t.
- α is the learning rate.
- m_t and v_t are the first and second moment estimates of the gradients, respectively.
- ϵ is a small constant to avoid division by zero.

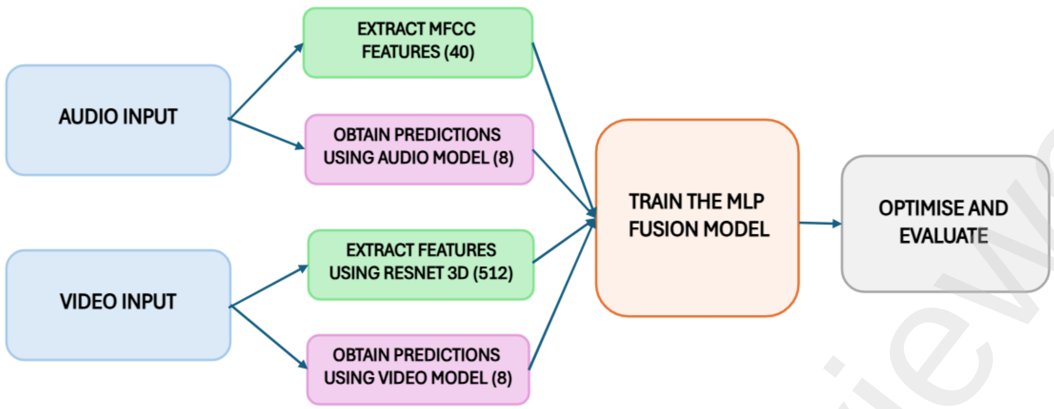

Fig. 5. Fusion Mechanism

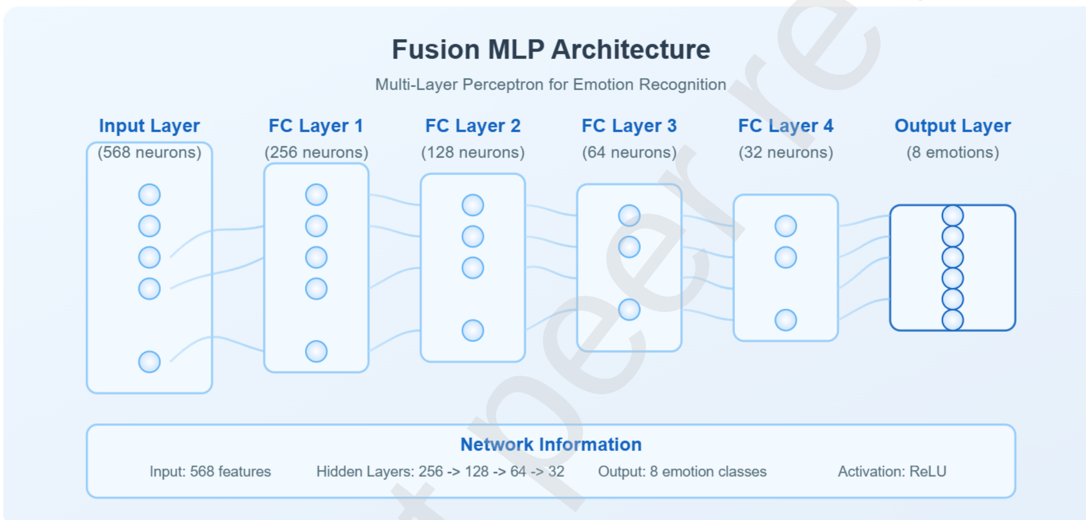

Fig. 6. Architecture of Fusion Model

Adam's adaptively scaling the learning rates renders it highly useful for training deep networks consisting of a mixture of convolutional, recurrent, and fully connected layers.

*3) **Regularization Techniques**:* To avoid overfitting and enhance generalization performance, some regularization methods are applied during training:

***Dropout Layers**:* Dropout is implemented in the fully connected layers to randomly shut off neurons while training, making the network learn general features and not memorize the patterns. It applies a 0.4 dropout rate in critical layers to curb overfitting but retain meaningful representations.

***Batch Normalization**:* Batch normalization is used to stabilize training and combat internal covariate shift by normalizing the activations in every layer. This helps to keep gradient updates stable so that they don't fluctuate wildly, which could hinder convergence.

***Early Stopping**:* To further reduce overfitting, early stopping is used. The model keeps track of the validation loss at all times, and if there is no improvement for a specified number of epochs, training is stopped to avoid unnecessary weight updates.

E. Hyperparameter Tuning

*1) **Learning Rate Scheduling**:* A learning rate scheduler is used to dynamically change the learning rate during training. A higher learning rate is initially used to enable quick weight adjustments, but as training continues, the learning rate is slowly decreased to enable fine-tuned weight updates.

The scheduler decreases the learning rate as validation loss plateaus to prevent the model from fluctuating about poor solutions.

*2) **Batch Size Selection**:* Selecting the best batch size is essential to balance training efficiency and generalization.

- Fast training is facilitated by bigger batch sizes but could result in poor generalization through lower noise in weight updates.

- Smaller batch sizes improve generalization but are more time-consuming to train.

A grid search method is employed to find the best batch size by testing training speed and validation accuracy on various configurations.

*3) **Number of Layers and Units**:* The model architecture, i.e., the count of CNN filters, LSTM units, and MLP neurons, is optimized by hyperparameter tuning methods like:

***Grid Search**:* Evaluates predefined parameter combinations.

***Random Search**:* Selects hyperparameters randomly from a specified range, enhancing exploration.

The goal is to optimize performance while staying efficient.

F. Cross-Validation Strategy

To make the model generalize correctly for different distributions of data, five-fold cross-validation is adopted. The database is split into five subsets wherein:

- Four subsets are used for training.
- One subset is used for validation.

This is done five times so that each sample is utilized for both training and validation at least once. Cross-validation avoids overfitting by testing the model on various subsets of the dataset, making it less dependent on individual training instances.

V. Evaluation and Testing

***Evaluation Metrics**:* In order to measure the performance of the suggested multimodal model, several performance measurements are employed.

*1) **Accuracy**:* Accuracy measures the overall correctness of the model's predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

where:

- TP (True Positives) are properly predicted emotional occurrences.
- TN (True Negatives) are accurately predicted non-emotional instances.
- FP (False Positives) are wrongly labeled with emotions.
- FN (False Negatives) are true feelings that were missed.

*2) **Precision**:* Precision evaluates how many emotion predictions were correct among all predictions made for that class:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{12}$$

A high precision score indicates that the model minimizes false positives, which is especially important in applications where misclassification of emotions could lead to incorrect decisions.

*3) **Recall (Sensitivity)**:* Recall measures the model's ability to correctly identify all actual instances of a specific emotion:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{13}$$

A low recall score means that the model is missing a significant number of emotional expressions, leading to high false negatives.

*4) **F1-Score**:* The F1-score is the harmonic mean of precision and recall, ensuring a balance between these two metrics:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

A high F1-score means that the model maintains both high precision and high recall, making it a robust metric for imbalanced datasets where some emotions appear more frequently than others.

VI. Dataset Description

The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) is the major dataset used for training and testing the multimodal emotion recognition system, consisting of audio and video records of 24 actors performing eight emotions in a controlled environment. The dataset offers a standardized test bench for emotion classification based on speech and facial expressions.

*1) **Structure of the RAVDESS Dataset**:* The dataset contains emotion-tagged audio and video samples. The audio recordings are in '.wav' format, whereas the video recordings are in '.mp4' format. The actors deliver two predefined speech statements with different emotional strengths, thereby presenting a variety of emotions.

*2) **Emotions Included in the Dataset**:* The database contains recordings of eight emotion categories: neutral, calm, happy, sad, angry, fearful, disgusted, and surprised. All actors use the same expressions, which have a balanced split across all classes of emotions.

*3) **Audio Data**:* The audio samples consist of speech and singing recordings, allowing for the analysis of intonation, pitch, and rhythm variations in emotional speech. They providing high-fidelity signals suitable for deep learning-based feature extraction.

*4) **Video Data**:* The video samples record facial expressions that match each emotion, enabling spatial and temporal analysis of emotional signals. The dataset contains both full audio-visual (AV) recordings and video-only samples to provide flexibility in modality choice.

*5) **Naming Convention of Files**:* Each file follows a structured naming format.

For example, '03-01-01-01-02-02-03.wav' represents:

- Modality: (01 = full-AV, 02 = video-only, 03 = audio-only).

- Vocal channel: (01 = speech, 02 = song).
- Emotion: (e.g., 01 = neutral).
- Emotional intensity: (01 = normal, 02 = strong).
- Statement: (01 = "Kids are talking by the door", 02 = "Dogs are sitting by the door").
- Repetition: (01 = 1st repetition, 02 = 2nd repetition).
- Actor: (01 to 24. Odd-numbered actors are male, even-numbered actors are female).

*6) **Dataset Partitioning for Fusion Model:*** For model training of multimodal fusion, audio and video samples were paired to have consistent emotional expression. The dataset has 1,360 distinct samples, each with both an audio and video recording. The dataset was split into five disjoint folds, without any repetition within folds. This five-fold cross-validation approach provides a balanced distribution of emotions and avoids overfitting, enhancing model generalization.

VII. Algorithms and Technologies Used

The suggested multimodal emotion recognition system combines deep learning-based audio and video processing models, utilizing TensorFlow, PyTorch, and Scikit-Learn for feature extraction, training, and testing.

*1) **Audio Processing Pipeline:*** The audio model uses Mel-Frequency Cepstral Coefficients (MFCCs) as the main feature extraction technique, in addition to a hybrid deep learning architecture that includes CNN, BiLSTM, and SimpleRNN layers.

- Convolutional layers extract local spectral features in the speech signal.
- BiLSTM layers extract long-term temporal dependencies.
- RNN layers fine-tune sequential information to improve emotional representation.

The model is trained with the Adam optimizer, including dropout layers and batch normalization to avoid overfitting. Synthetic Minority Over-Sampling Technique (SMOTE) is used to address imbalanced emotion classes, providing a better-balanced dataset for training.

*2) **Video Processing Pipeline:*** For emotion recognition using video as input, the model employs a 3D Convolutional Neural Network (3D CNN) from ResNet3D (r3d_18) as the base model. Unlike 2D CNNs, which extract only spatial information, 3D CNNs handle both spatial and temporal characteristics in video frames, enabling the model to interpret facial expressions dynamically across time.

To further refine extracted features, additional 3D convolutional layers are stacked, followed by global average pooling for dimension reduction. The Adam optimizer is used with a learning rate of 0.0001, and batch normalization is applied to stabilize training.

*3) **Fusion Model for Multimodal Emotion Recognition:*** The fusion model integrates both the audio and video feature representations for improved classification performance.

- The audio feature vector consists of 40 MFCCs.
- The video feature vector is a 512-dimensional ResNet3D representation.

These vectors are concatenated with their respective Softmax probability distributions, resulting in a 568-dimensional multimodal feature representation.

This multimodal representation is then fed into a Multi-Layer Perceptron (MLP) classifier with four fully connected layers, each regularized using dropout and batch normalization.

To ensure optimized training, the Adam optimizer is used with weight decay, and a StepLR scheduler dynamically adjusts the learning rate over the epochs, enhancing model convergence.

*4) **Hyperparameter Tuning for Audio, Video, and Fusion Models:*** The precise hyperparameters used in the audio, video, and fusion models are presented in the ensuing tables.

A. Hyperparameters for the Audio Model

The audio model utilizes a sequential pipeline that includes feature extraction, convolutional processing, temporal modeling, and classification. The CNN-BiLSTM-RNN structure supports both the local extraction of features and sequential pattern learning from long distances and is therefore optimally suitable for speech-based emotion recognition. Training parameters employed in optimizing the model are given in Table IV.

B. Hyperparameters for the Video Model

The video-based model utilizes ResNet3D (r3d_18) as the feature extractor, which is further fine-tuned by adding additional 3D convolutional layers to improve spatial-temporal feature learning. The model records facial expressions changing dynamically in real time, enabling a strong emotion classification system. The training parameters utilized for video-based feature extraction and classification are listed in Table V.

C. Hyperparameters for the Fusion Model

The fusion model exploits both feature-based fusion and prediction-based fusion by pairing raw feature vectors with Softmax probability distributions. The 40 MFCC features extracted from the audio model, the 512-dimensional video feature vector, and the 8 probability values across each modality are concatenated into a 568-dimensional multimodal representation. This feature representation is then fed into a fully connected MLP classifier, optimized with the Adam optimizer and StepLR learning rate scheduler.

The combination of audio and video modalities guarantees that the model uses both speech and facial expressions to categorize emotions with greater precision. The fusion model takes advantage of the complementary nature of these modalities, as audio captures emotional tone and intensity, whereas video provides visual cues from facial expressions.

In general, the model is trained with a highly optimized set of hyperparameters, chosen empirically using grid search and evaluation methods to produce the best performance. Stable

TABLE IV
HYPERPARAMETERS FOR THE AUDIO MODEL

Parameter	Value
Learning Rate	0.0005
Batch Size	16
Epochs	100
Dropout Rate	0.4 (LSTM layers), 0.3 (Dense layer)
L2 Regularization	0.01
Optimizer	Adam
CNN Filters	64, 128, 256, 512 (4 Conv1D layers)
Kernel Size	3 (for Conv1D)
Pooling	MaxPooling1D (pool size = 2)
LSTM Units	256 (first BiLSTM), 512 (second BiLSTM)
RNN Units	512 (SimpleRNN layer)
Loss Function	Sparse Categorical Cross-Entropy
Learning Rate Scheduler	ReduceLROnPlateau (factor = 0.5, patience = 5, min_lr = 1e-6)
Data Augmentation	SMOTE (for handling imbalanced classes)

TABLE V
HYPERPARAMETERS FOR THE VIDEO MODEL

Parameter	Value
Learning Rate	0.0001
Batch Size	32
Epochs	15
Dropout Rate	0.4
Optimizer	Adam
Base Model	ResNet3D (r3d_18)
Additional Conv Layers	256, 128, 64 filters (kernel size = (3,3,3))
Pooling	AdaptiveAvgPool3d (Global Pooling)
Fully Connected Layers	32 (FC1), 8 (FC2, output layer)
Loss Function	Cross-Entropy Loss

TABLE VI
HYPERPARAMETERS FOR THE FUSION MODEL

Parameter	Value
Learning Rate	0.0005
Epochs	1000
Dropout Rate	0.5
Optimizer	Adam (with weight decay 1e-5)
Learning Rate Scheduler	StepLR (step_size = 100, gamma = 0.5)
MLP Layers	256, 128, 64, 32
Loss Function	Cross-Entropy Loss

training and generalization are guaranteed by the usage of dropout, batch normalization, and learning rate scheduling, with cross-validation being used to evaluate model robustness over varied data distributions.

VIII. RESULTS

This section discusses the performance analysis of the proposed emotion recognition system under three distinct modalities: audio only, video only, and multimodal fusion. Five-fold cross-validation was used to validate the stability and universal applicability of the model. The accuracy value for each fold and the mean of the overall accuracy for each modality is compiled in Table VII.

The findings show that the audio-only model had a mean accuracy of 79.24%. This implies that vocal features like tone, pitch, and speech rhythm play an important role in emotion recognition. Nevertheless, some emotions, especially those with similar acoustic features like neutral and sadness, were challenging to classify. The performance of the audio-based model could also have been influenced by background noise, speech intensity variations, and speaker-dependent issues.

The video-only scheme achieved a significantly superior mean accuracy of 92.72%, suggesting that facial expressions yield a more discriminable and consistent source of information for emotion classification. The fluctuation of accuracy values from 88.2% to 96.6% indicates that there could be certain emotions with very subtle or equivocal visual manifestations, resulting in misclassifications. Aspects like lighting conditions, occlusion, and diversity of facial expressions could have impacted the video-based system performance.

The multimodal fusion model, which combined both audio and video features, had the best mean accuracy of 96.06%. This remarkable improvement illustrates that the combination of speech and facial information improves the overall classification performance since the model can use complementary information from both modalities. The fusion model outperformed the unimodal models in all folds, demonstrating the benefits of using multiple data sources for emotion recognition. The decrease in performance variance over the various folds indicates that the fusion model is more stable and robust than the unimodal models.

The overall results indicate that video-based emotion recognition performs better than audio-based recognition, highlighting the importance of facial expressions as a primary mode of emotional expression. Nevertheless, the combination of both modalities yields the optimal performance, confirming that multimodal models are less ambiguous and more robust, especially when one modality alone might not be enough to accurately discriminate between emotions.

IX. COMPARATIVE ANALYSIS OF FUSION ACCURACY

The proposed emotion recognition system achieved a **mean fusion accuracy of 96.06%**, surpassing both the audio-only (79.24%) and video-only (92.72%) models. This sec-

TABLE VII
PERFORMANCE COMPARISON ACROSS MODALITIES

Fold	Audio Accuracy (%)	Video Accuracy (%)	Fusion Accuracy (%)
Fold 1	78.8	93.0	95.6
Fold 2	80.6	91.9	97.6
Fold 3	77.4	93.7	97.2
Fold 4	79.8	96.6	95.3
Fold 5	79.6	88.2	94.6
Mean Accuracy	**79.24**	**92.72**	**96.06**

tion presents a detailed comparison with existing literature, highlighting key findings and potential areas for further enhancement.

A. Fusion Accuracy from Other Studies

Fig 7 presents a comparison of fusion accuracy across various studies in multimodal emotioan recognition. The accuracy values illustrate the improvements brought about by different fusion techniques and architectural choices.

The results indicate that the proposed model outperforms all existing methods in terms of fusion accuracy. The closest competing method is the transformer-based multimodal fusion approach, which achieved 93.59% accuracy. This suggests that self-attention mechanisms and deep embeddings contribute significantly to feature alignment and fusion effectiveness. Traditional deep learning-based fusion approaches, such as cross-attentional fusion and multiplicative fusion, show slightly lower accuracy values of 89.25% and 89.00%, respectively. The difference in performance can likely be attributed to how well each method integrates complementary information from the audio and video streams.

B. Analysis of Model Performance and Robustness

A crucial aspect of multimodal emotion recognition is the stability of the model across different evaluation folds. The proposed system exhibits a lower variance across folds compared to other methods, indicating better generalization across diverse samples. The five-fold cross-validation strategy employed ensures that the model is rigorously tested against varying distributions of data, making the reported accuracy values more reliable.

Another important consideration is the role of modality-specific dependencies. The transformer-based approaches, as seen in prior works, appear to benefit from attention-based mechanisms that dynamically weigh the importance of features from different modalities. In contrast, simpler concatenation or additive fusion strategies may not be as effective in capturing complex interdependencies between audio and video signals. The superior performance of the proposed system suggests that its fusion strategy effectively mitigates redundancy while preserving the most discriminative features from each modality.

Furthermore, while video-based models consistently outperform audio-based models, it is evident that audio still contributes valuable information that enhances classification accuracy when combined with visual data. The fusion strategy used in the proposed system likely addresses the issue of feature complementarity more efficiently than prior methods, leading to its superior accuracy.

C. Potential Areas for Future Improvement

Despite achieving the highest fusion accuracy among the compared models, there remain opportunities for further refinement. Transformer-based fusion techniques continue to show promise, and exploring variations such as hierarchical attention mechanisms or multi-scale feature aggregation could yield even better performance. Additionally, incorporating domain adaptation techniques to enhance model robustness across different datasets may further improve generalization.

The impact of external factors such as background noise in audio or occlusions in video remains an open challenge in multimodal emotion recognition. Addressing these issues through adaptive preprocessing techniques or adversarial training could help make the system more robust in real-world applications.

X. CONCLUSION

The comparative analysis demonstrates that the proposed system achieves the **highest fusion accuracy (96.06%)** among existing methods. The superior performance can be attributed to an effective fusion strategy that captures complementary information from both modalities while minimizing redundancy. The findings suggest that transformer-based and deep metric learning approaches provide significant benefits for multimodal fusion. Future work may explore additional refinements, including enhanced attention mechanisms, adaptive feature selection, and domain adaptation techniques, to further push the boundaries of multimodal emotion recognition.

The research also established the superiority of deep learning-based fusion over probability-based fusion methods. The results indicate that a hybrid solution involving both speech and facial expression analysis is essential in developing a robust, real-time emotion recognition system. Future enhancements may lie in the creation of larger datasets, utilization of transformer-based architectures, and establishment of real-time implementations, laying the ground for sophisticated applications in human-computer interaction, affective computing, and mental health monitoring.

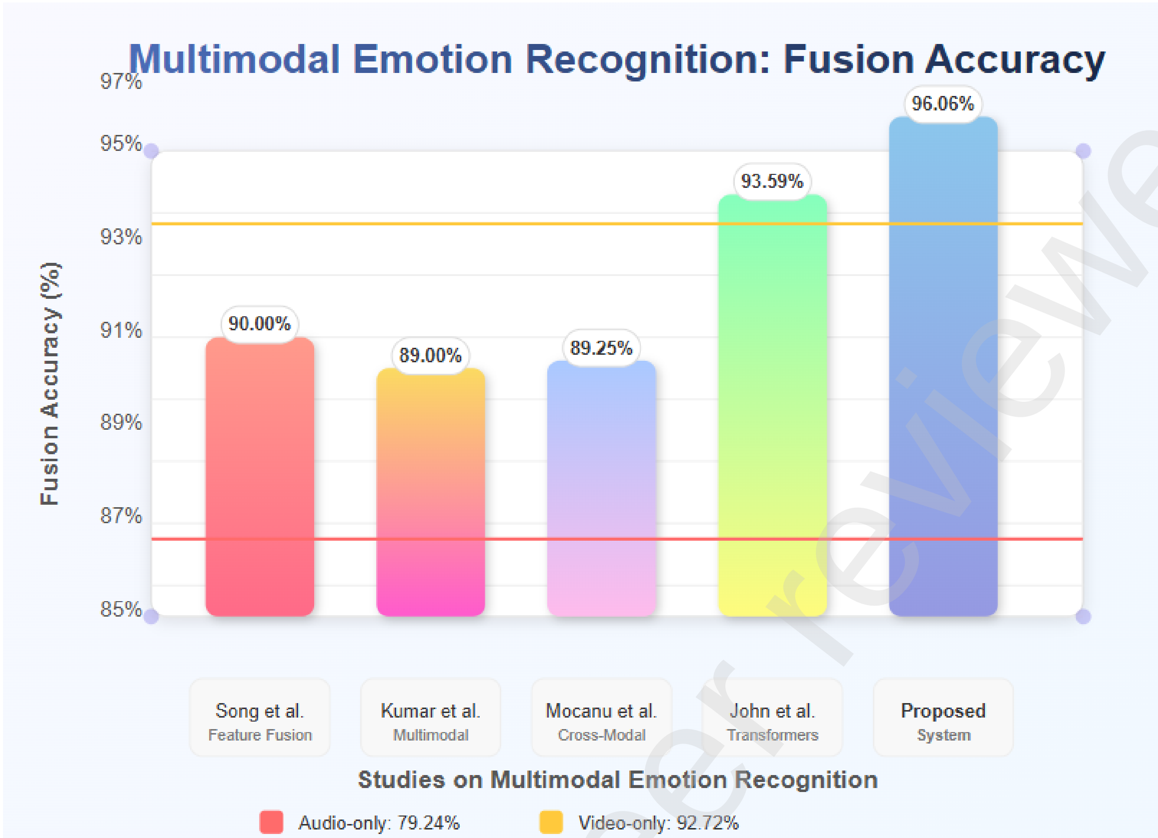

Fig. 7. Comparitive Analysis of the performance for RAVDESS dataset

References

[1] B. Mocanu, R. Tapu, and T. Zaharia, "Multimodal emotion recognition using cross-modal audio-video fusion with attention and deep metric learning," *Image Vis. Comput.*, vol. 133, p. 104676, 2023. doi: 10.1016/j.imavis.2023.104676.

[2] M. Z. Uddin and E. G. Nilsson, "Emotion recognition using speech and neural structured learning to facilitate edge intelligence," *Eng. Appl. Artif. Intell.*, vol. 94, p. 103775, 2020. doi: 10.1016/j.engappai.2020.103775.

[3] S. Zhang, Y. Yang, C. Chen, X. Zhang, Q. Leng, and X. Zhao, "Deep learning-based multimodal emotion recognition from audio, visual, and text modalities: A systematic review of recent advancements and future prospects," *Expert Syst. Appl.*, vol. 237, p. 121692, 2024. doi: 10.1016/j.eswa.2023.121692.

[4] M. Singh and Y. Fang, "Emotion recognition in audio and video using deep neural networks," Stanford Univ., Stanford, CA, USA, [Online], 2020.

[5] V. John and Y. Kawanishi, "Audio and video-based emotion recognition using multimodal transformers," in *Proc. 26th Int. Conf. Pattern Recognit. (ICPR)*, Montréal, QC, Canada, Aug. 2022.

[6] A. Shrivastava, D. Dubey, M. Verma, and H. Verma, "Facial emotion recognition using video and audio," *Int. J. Res. Publ. Rev.*, vol. 5, no. 1, pp. 2517-2527, Jan. 2024. doi: 10.55248/gengpi.5.0124.0261.

[7] M. S. Hossain and G. Muhammad, "Emotion recognition using deep learning approach from audio-visual emotional big data," *Inf. Fusion*, vol. 49, pp. 69-78, Sep. 2018. doi: 10.1016/j.inffus.2018.09.008.

[8] H. Kumar and A. Martin, "Multimodal approach: Emotion recognition from audio and video modality," in *Proc. 7th Int. Conf. Electron., Commun. Aerosp. Technol. (ICECA)*, Thiruvarur, India, 2023.

[9] E. Cambria, X. Zhang, R. Mao, M. Chen, and K. Kwok, "SenticNet 8: Fusing emotion AI and commonsense AI for interpretable, trustworthy, and explainable affective computing," *IEEE Transactions on Affective Computing*, 2023.

[10] Y. Song, Y. Cai, and L. Tan, "Video-audio emotion recognition based on feature fusion deep learning method," Indiana, USA, 2021.

[11] J. Han, Z. Zhang, Z. Ren, and B. Schuller, "EmoBed: Strengthening Monomodal Emotion Recognition via Training with Crossmodal Emotion Embeddings," *IEEE Trans. Affective Comput.*, vol. 12, no. 3, pp. 553–567, 2019. doi: 10.1109/TAFFC.2019.2928297.

[12] S. Wang, G. Peng, Z. Zheng, and Z. Xu, "Capturing Emotion Distribution for Multimedia Emotion Tagging," *IEEE Trans. Affective Comput.*, vol. 12, no. 4, pp. 821–835, 2021. doi: 10.1109/TAFFC.2019.2900240.

[13] J. Deng and F. Ren, "Multi-Label Emotion Detection via Emotion-Specified Feature Extraction and Emotion Correlation Learning," *IEEE Trans. Affective Comput.*, vol. 14, no. 1, pp. 475–489, 2023. doi: 10.1109/TAFFC.2020.3034215.

[14] S. Kamada and T. Ichimura, "Multimodal Adaptive Structural Deep Belief Network for Emotion Recognition on RAVDESS Dataset," in *Proc. 16th IIAI Int. Congr. Adv. Appl. Inform. (IIAI-AAI)*, 2024. doi: 10.1109/IIAI-AAI63651.2024.00062.

[15] A. Li, M. Wu, R. Ouyang, Y. Wang, F. Li, and Z. Lv, "A Multimodal-Driven Fusion Data Augmentation Framework for Emotion Recognition," *IEEE Trans. Artif. Intell.*, 2025. doi: 10.1109/TAI.2025.3537965.